# Symmetries in Physics — Exercise Sheet 8

*Tensor products, selection rules and level splitting*

**Variant: Questions only**

**Exercise 1.** The symmetry group $D_4$ of the square has the character table (Sheet 5, Exercise 2)

| $D_4$ | $1\,e$ | $1\,r^2$ | $2\,r$ | $2\,s$ | $2\,rs$ |
|---|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 | 1 |
| $A_2$ | 1 | 1 | 1 | −1 | −1 |
| $B_1$ | 1 | 1 | −1 | 1 | −1 |
| $B_2$ | 1 | 1 | −1 | −1 | 1 |
| $E$ | 2 | −2 | 0 | 0 | 0 |

The two-dimensional representation $E$ of $D_4$ acts on vectors $\vec{a}$ in the plane of the square by the matrices

$$D(r) = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

the quarter turn and the reflection in the $x$-axis.

(i) Decompose $E \otimes E$ by characters.

(ii) Split it into $S^2E$ and $\Lambda^2E$ with the formula for the symmetric and antisymmetric squares (Lecture 8).

(iii) Let $\vec{a}$ and $\vec{b}$ transform by $E$. For each type found in (ii), give the combination of the products $a_\alpha b_\beta$ that carries it, as Lecture 8 does for $C_{3v}$. Why does the pair $(a_x b_x - a_y b_y,\ a_x b_y + a_y b_x)$, which carries $E$ in $C_{3v}$, not stay together here?

**Exercise 2.** In a $C_{3v}$-symmetric system an operator $(T_x, T_y)$ of type $E$ acts between two doublet levels $\mathcal{L}_i$ and $\mathcal{L}_f$, both of type $E$. The bases $|i_\nu\rangle$, $|f_\mu\rangle$ of the levels and the pair $(T_x, T_y)$ all transform by the matrices of Lecture 5, with hydrogen 1 on the $y$-axis:

$$D(C_3) = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad D(\sigma_1) = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

(i) Explain why the eight amplitudes $\langle f_\mu | T_\alpha | i_\nu \rangle$, $\mu, \alpha, \nu \in \{x, y\}$, are fixed by a single reduced matrix element.

(ii) Find the intertwiner $C\colon E \otimes E \to E$ by imposing $C\big(D(g) \otimes D(g)\big) = D(g)\,C$ for $g = \sigma_1$ and $g = C_3$.

(iii) Which amplitudes vanish, and how are the others related?

(iv) Show that the cubic $3x^2y - y^3$ is invariant under $C_{3v}$, and compare its coefficients with (iii).

**Exercise 3.** In a $C_{3v}$-symmetric system the electric dipole operator has the component $z$ of type $A_1$ and the pair $(x, y)$ of type $E$ (Lecture 5).

(i) Use the selection rule of Lecture 8 to list the transitions between levels of types $A_1$, $A_2$ and $E$ that light polarised along the axis ($z$) can cause, and those that light polarised in the plane $(x, y)$ can cause.

(ii) Show that $A_1 \leftrightarrow A_2$ is forbidden for every polarisation.

(iii) What type must an operator have to connect an $A_1$ level with an $A_2$ level? Give a physical example.

**Exercise 4 (formaldehyde in the infrared and in Raman scattering).** Sheet 7, Exercise 2 placed formaldehyde $H_2CO$ in the $yz$-plane, with the C–O bond along the $z$ axis, and found $\Gamma_{\text{vib}} = 3A_1 \oplus 2B_1 \oplus B_2$ with the $C_{2v}$ table

| $C_{2v}$ | $E$ | $C_2$ | $\sigma_v(yz)$ | $\sigma'_v(xz)$ | |
|---|---|---|---|---|---|
| $A_1$ | 1 | 1 | 1 | 1 | $z$ |
| $A_2$ | 1 | 1 | −1 | −1 | |
| $B_1$ | 1 | −1 | 1 | −1 | $y$ |
| $B_2$ | 1 | −1 | −1 | 1 | $x$ |

(i) Which fundamentals can be infrared active, and along which axis does the dipole of each oscillate? Use Lecture 8's argument: the matrix $L = \partial\vec{d}/\partial q$ at $q = 0$ satisfies $L\rho(g) = R(g)L$.

(ii) Compute the character of $S^2\Gamma_{\text{vec}}$ and decompose it. Which fundamentals can be Raman active?

(iii) Which type would be Raman active but silent in the infrared? Does formaldehyde have a mode of that type?

**Exercise 5 (mutual exclusion rule).**

(i) Let the point group of a molecule contain the inversion $i$. Show that $\rho(i) = \pm\mathbb{1}$ in every irreducible representation, so that each irreducible has a definite parity: $g$ (*gerade*) for $\rho(i) = +\mathbb{1}$ and $u$ (*ungerade*) for $\rho(i) = -\mathbb{1}$. Then show that a $u$ mode can only be infrared active and a $g$ mode only Raman active. Lecture 8 quotes this step for its mutual exclusion rule.

(ii) Boron trifluoride has $\Gamma_{\text{vib}} = A'_1 \oplus 2E' \oplus A''_2$ (Sheet 7, Exercise 5). Find $\Gamma_{\text{vec}}$ and $S^2\Gamma_{\text{vec}}$. Decide which modes can be infrared active and which Raman active. Give the polarisation each infrared-active mode absorbs. Say why the mutual exclusion rule does not apply.

*Data.* The character table of $D_{3h}$, with classes of sizes 1, 2, 3, 1, 2, 3, is

| $D_{3h}$ | $E$ | $2C_3$ | $3C_2$ | $\sigma_h$ | $2S_3$ | $3\sigma_v$ |
|---|---|---|---|---|---|---|
| $A'_1$ | 1 | 1 | 1 | 1 | 1 | 1 |
| $A'_2$ | 1 | 1 | −1 | 1 | 1 | −1 |
| $E'$ | 2 | −1 | 0 | 2 | −1 | 0 |
| $A''_1$ | 1 | 1 | 1 | −1 | −1 | −1 |
| $A''_2$ | 1 | 1 | −1 | −1 | −1 | 1 |
| $E''$ | 2 | −1 | 0 | −2 | 1 | 0 |

The $C_3$ axis is $z$, perpendicular to the molecular plane $\sigma_h$.

**Exercise 6.** Stretch a square slightly along the $x$-axis into a rectangle. The symmetry drops from $D_4$ to its subgroup $D_2 = \{e, r^2, s, r^2s\}$, with $s$ the reflection in the $x$-axis as in Exercise 1. Restrict each irreducible representation of $D_4$ (table in Exercise 1) to $D_2$. Show that a doublet $E$ of the square generically splits into two nondegenerate levels of the rectangle.

**Exercise 7 (a tetragonal crystal field).** In Lecture 8's self-study an octahedral site splits a $d$ level into $E \oplus T_2$. Now stretch the octahedron slightly along the $z$ axis.

(i) The rotations that survive form a group isomorphic to $D_4$, with $r$ the quarter turn about $z$ and $s$ the half-turn about $x$. List its eight rotations, and find the class of $O$ that each one belongs to. (In Lecture 8's table of $O$, $3C_2$ are the half-turns about the coordinate axes, and $6C_2'$ the half-turns about the axes through the midpoints of opposite edges.)

(ii) Restrict $E$ and $T_2$ to $D_4$ and decompose (table in Exercise 1). How does the $d$ level split, generically, under a small stretch? Which of the orbitals $z^2$, $x^2 - y^2$, $xy$, $xz$, $yz$ goes with which type?

(iii) Back in the undistorted octahedron, use Lecture 8's rotation character formula to show that a $p$ level does not split, and that an $f$ level splits as $A_2 \oplus T_1 \oplus T_2$.